Linear Regression Program

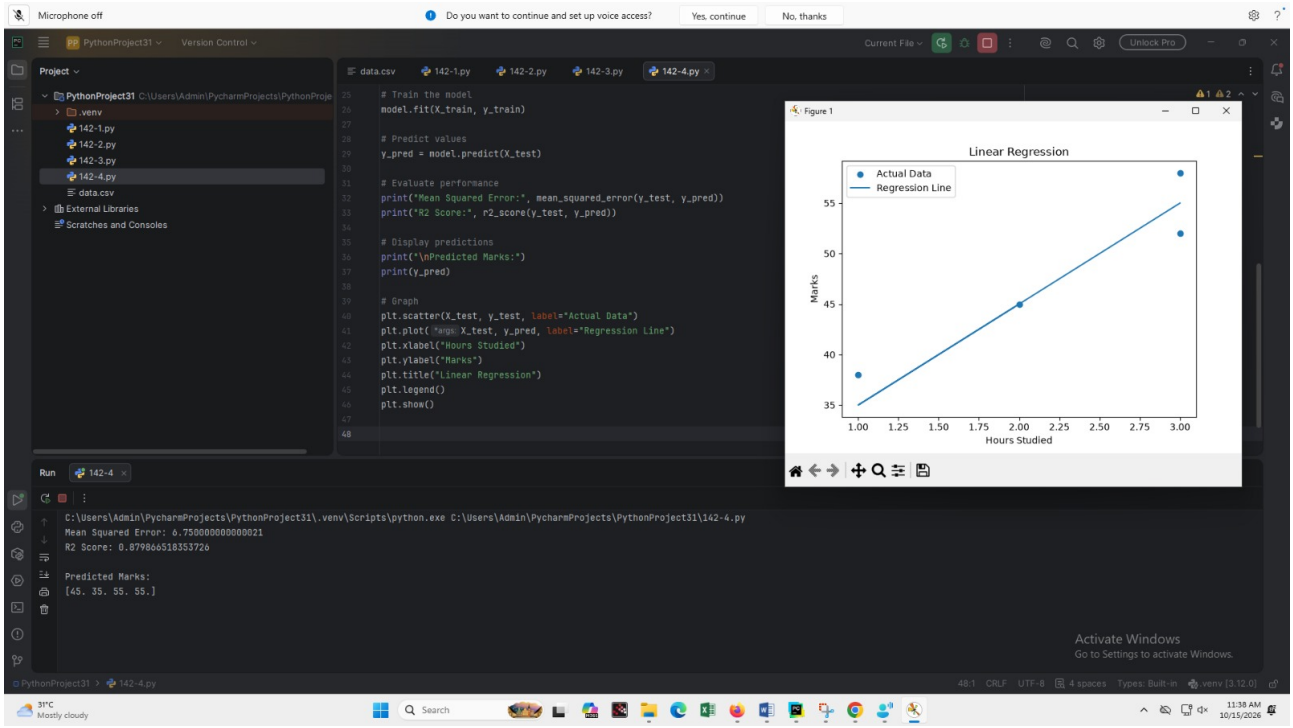
Aim

To write a program that builds a linear regression model on the given dataset, trains it on a training set, predicts values for the test set, and evaluates how well the fitted line explains the data using error metrics and the R^2 score.

Algorithm

1. Start the program.
2. Import the required libraries (pandas, numpy, matplotlib, train_test_split from sklearn.model_selection, LinearRegression from sklearn.linear_model, and mean_squared_error and r2_score from sklearn.metrics).
3. Load the dataset from the input file.
4. Separate the data into the input feature(s) X and the target variable y.
5. Split the data into training and testing sets using train_test_split (for example 80% training and 20% testing), with a fixed random_state.
6. Create the LinearRegression model object.
7. Train the model by fitting it on the training data (X_train, y_train).
8. Display the learned slope (coefficient) and intercept.
9. Predict the target values for the test set (X_test).
10. Calculate the Mean Squared Error and the R^2 score by comparing the predicted values with the actual values (y_test).
11. Display the predicted values, the MSE and the R^2 score.
12. Plot the actual data points and the fitted regression line (optional).
13. Stop the program.

Program


Microphone off
Do you want to continue and set up voice access?
Yes, continue
No, thanks
PythonProject31
Version Control
Current File
Unlock Pro
Project
PythonProject31 C:\Users\Admin\PycharmProjects\PythonProje
.venv
142-1.py
142-2.py
142-3.py
142-4.py
data.csv
External Libraries
Scratches and Consoles
data.csv
142-1.py
142-2.py
142-3.py
142-4.py
Train the model
model.fit(X_train, y_train)
Predict values
y_pred = model.predict(X_test)
Evaluate performance
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
Display predictions
print("\nPredicted Marks:")
print(y_pred)
Graph
plt.scatter(X_test, y_test, label="Actual Data")
plt.plot(*args: X_test, y_pred, label="Regression Line")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Linear Regression")
plt.legend()
plt.show()
Figure 1
Linear Regression
Actual Data
Regression Line
Marks
Hours Studied
55
50
45
40
35
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Run
142-4
C:\Users\Admin\PycharmProjects\PythonProject31\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject31\142-4.py
Mean Squared Error: 0.750000000000021
R2 Score: 0.879866518353726
Predicted Marks:
[45. 35. 55. 55.]
Activate Windows
Go to Settings to activate Windows.
PythonProject31 > 142-4.py
48:1
CRLF
UTF-8
4 spaces
Types: Built-in
.venv (3.12.0)
31°C
Mostly cloudy
Search
11:38 AM
10/15/2026